



Universidad
LOYOLA

Gestión del Conocimiento con Agentes IA Aplicación en Investigación

Seminarios Escuela de Ingeniería 2026-2027

ÍNDICE

1

Second Brain. Fundamentos

2

Recursos Prácticos

3

Aplicación a Investigación Doctoral

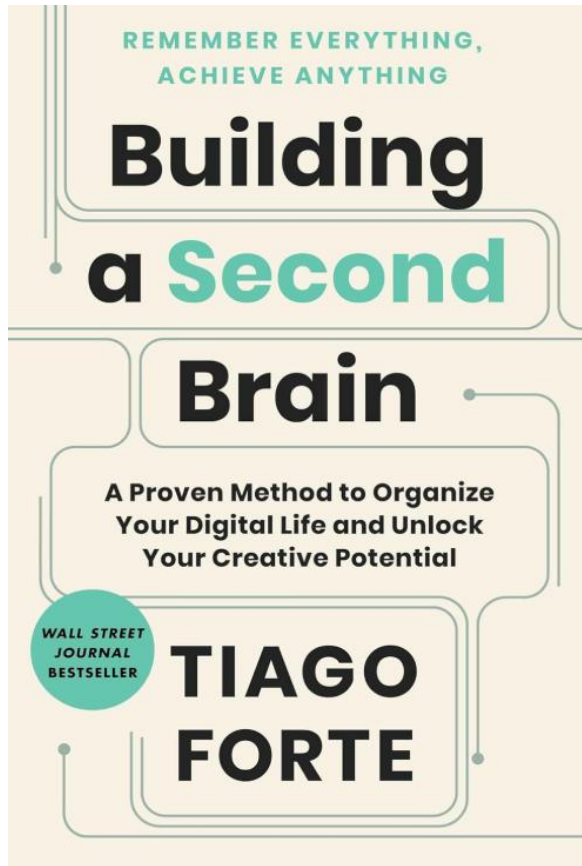
4

Caso de Uso:
Auditoría de Investigación
Asistida por IA

Second Brain Fundamentos



Second Brain: Tiago Forte



- Tiago Forte popularizó la metodología en el bestseller de 2022
- Fundó la consultora [Forte Labs](#)
- Expandió su metodología con el lanzamiento de *The AI Second Brain*, un programa diseñado para estimular la creatividad de manera estructurada

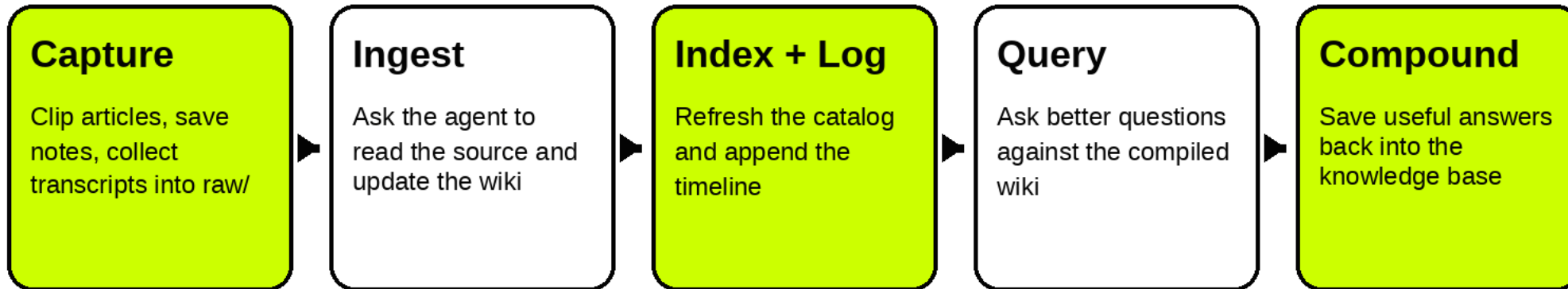
LLM Wiki Pattern: Andrej Karpathy



**“Obsidian is the IDE;
the LLM is the programmer;
the wiki is the codebase.”**

— Andrej Karpathy, [LLM Wiki \(2026\)](#)

Proceso de compilación y consulta



“Supera” al RAG común:

- **Precomputación del conocimiento.** En lugar de trabajar sobre *chunks*, las queries se hacen a sobre un grafo, nodos conectado por *wiki-links*
 - *La IA mantiene una Wiki ya digerida, sintetizada y cruzada*
- **Legible por humanos:** La IA escribe la Wiki en archivos de texto plano **Markdown** (.md)

Recursos Prácticos



Autogenerable como Agente Claude

El proceso encadena tres skills acoplados por fichero (ninguno lee la salida en memoria del otro; se comunican a través de artefactos en disco, lo que lo hace reanudable y auditable):

Skill	Rol	Cadencia
obsidian-vault-builder	crea el vault y su config <code>.obsidian/</code> sin abrir Obsidian	bootstrap (una vez)
karpathy-llm-wiki	ingiere fuentes, responde consultas, hace lint; orquesta a los otros dos	bucle (por fuente)
obsidian-graph-colors	colorea el Graph view por los tags del frontmatter	mantenimiento

`karpathy-llm-wiki` es el punto de entrada: delega el bootstrap en `obsidian-vault-builder` y el coloreado en `obsidian-graph-colors`.

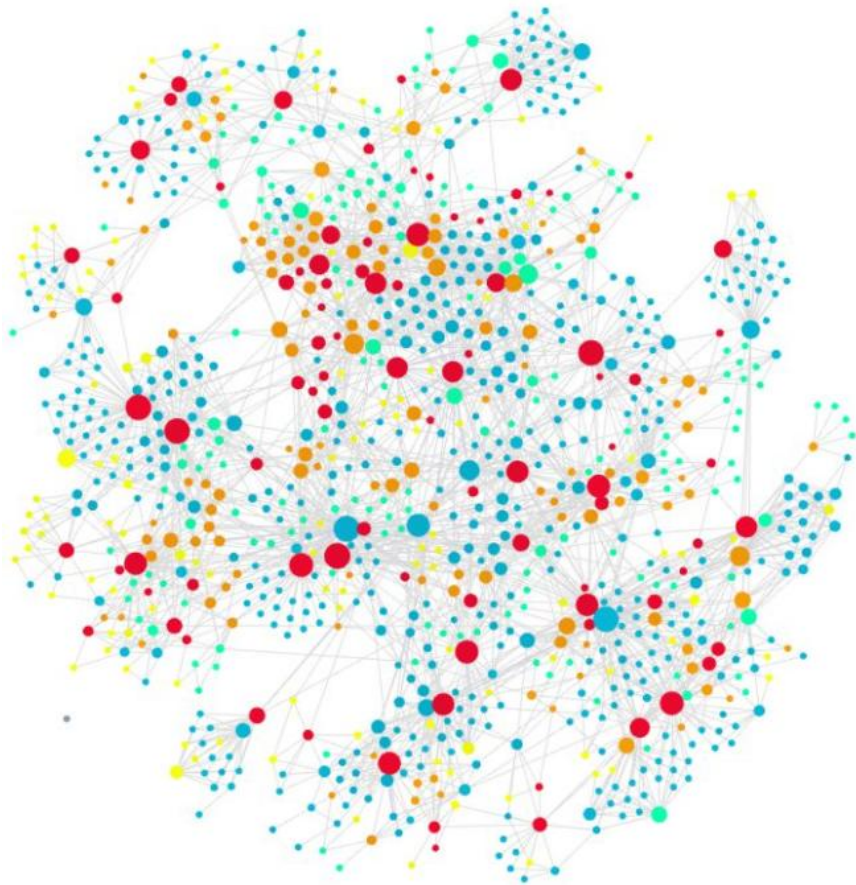


[@therobotacademy/obsidian-llm-wiki](https://github.com/@therobotacademy/obsidian-llm-wiki)

Arquitectura en cuatro capas

```
corpus de fuentes (PDF/DOCX/web/notas)  🔒 immutable, read-only
  |  extracción a texto (p.ej. pdfplumber)
  ▼
raw/<topic>/YYYY-MM-DD-slug.md           🔒 immutable una vez creado
  |  compile (merge / new / cross-topic)
  ▼
wiki/<topic>/articulo.md + index.md + log.md  ← propiedad del LLM, «compounding»
  |
  ▼
wiki/.obsidian/graph.json                 ← config (⚠ editar con Obsidian cerrado)
```

Aplicación a Bases de Código



Grupos	
tag:#code/class	Orange
tag:#code/function	Blue
tag:#code/method	Green
tag:#code/file	Red
tag:#code/variable	Yellow
tag:#index	Grey

Un pasito más:

“¿Y si lo extendemos al problema de la documentación de código?”



[@therobotacademy/Codegraph-to-Obsidian](https://github.com/therobotacademy/Codegraph-to-Obsidian)

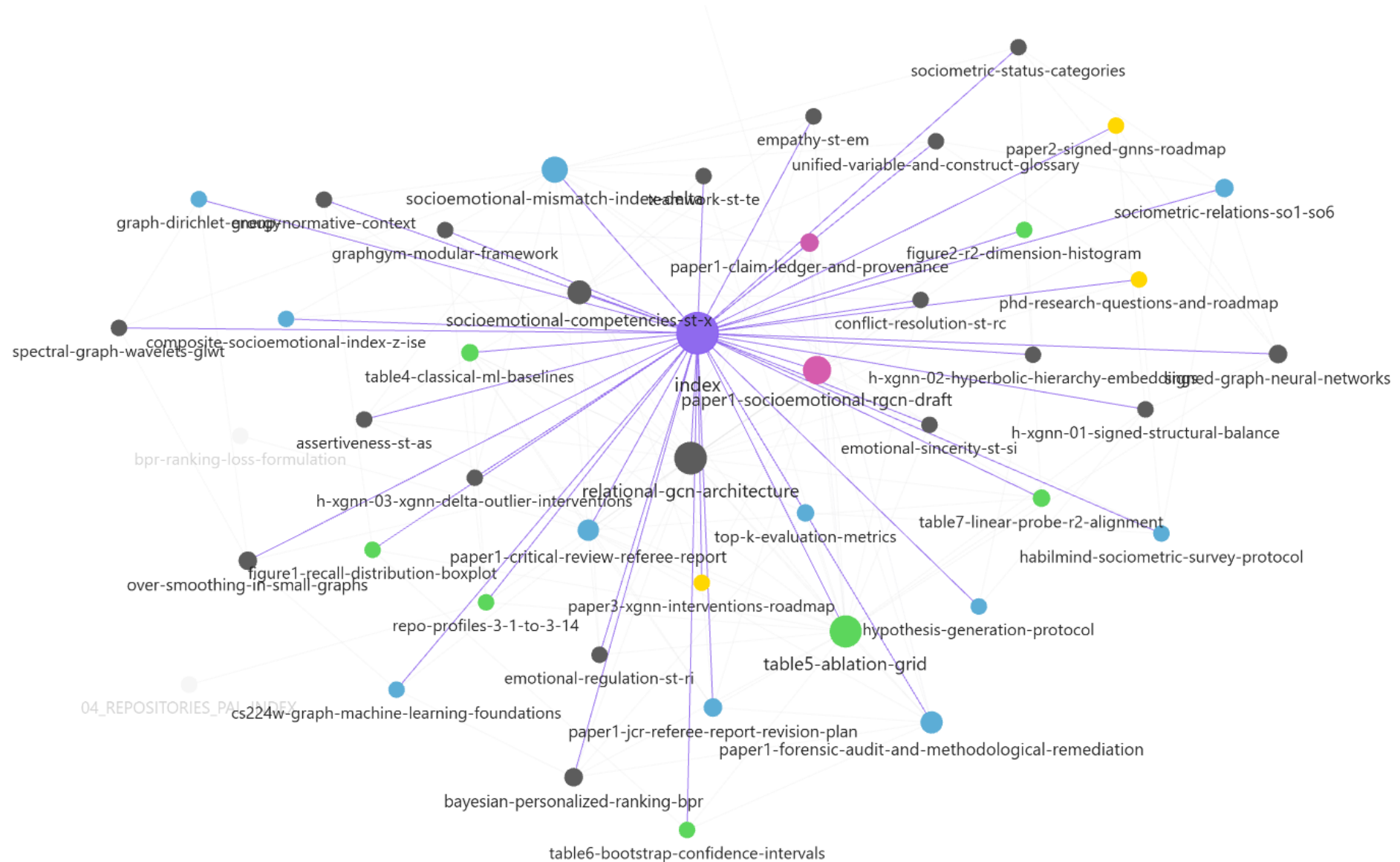
Aplicación a Investigación Doctoral



10. *Journal of the American Medical Association*, 2000; 284: 1039-1044.



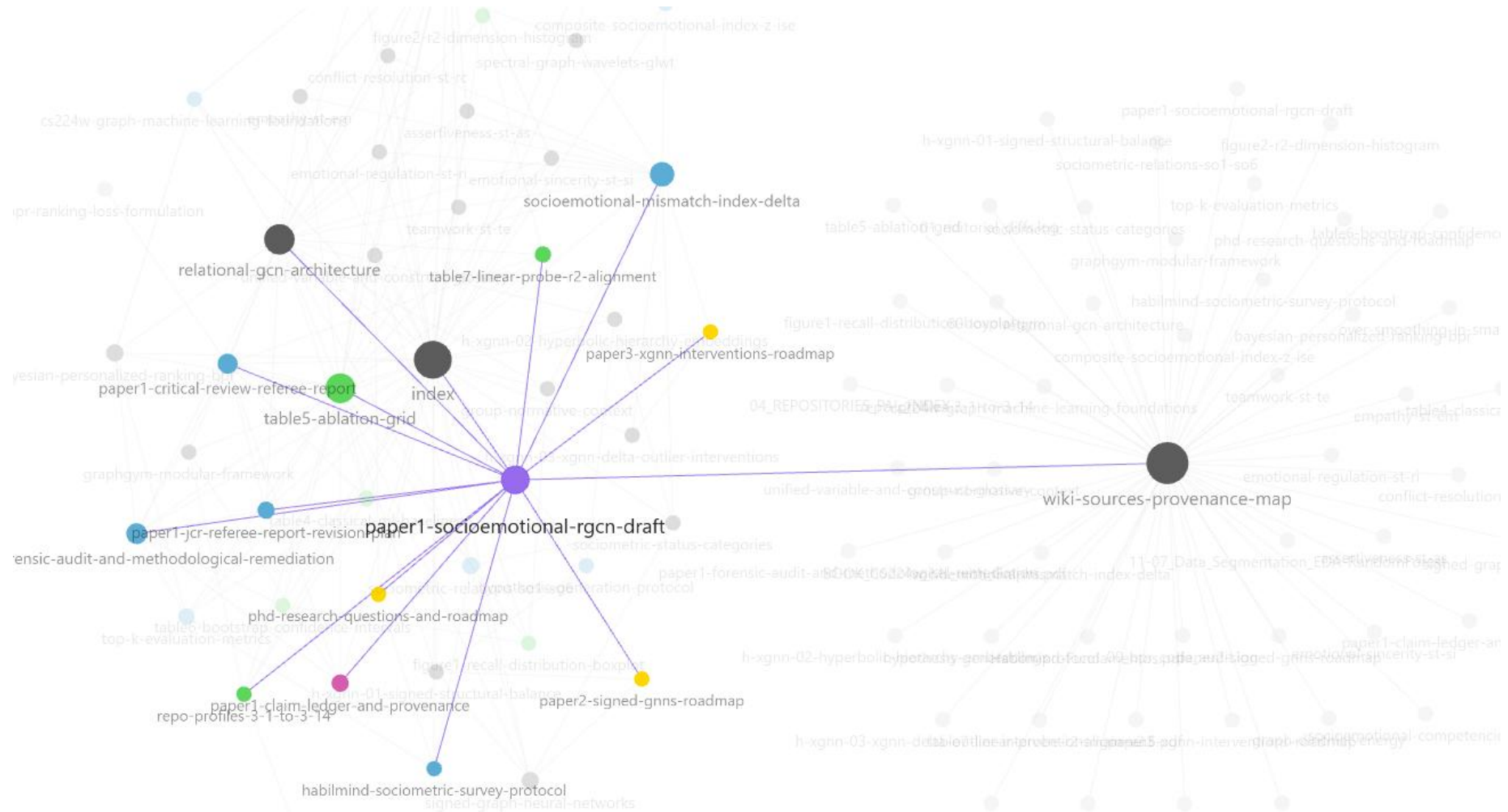
1. *Journal of the American Medical Association*, 2000; 284: 2689-2695.







Trazabilidad del paper



Wiki: Thesis-Meta / Theory Psychology

- > 00-thesis-meta
- > 01-theory-psychology
- > 02-theory-gnn
- > 03-methodology
- > 04-experiments-code
- > 05-papers-publications
- > 06-hypotheses
- index
- log

- ▼ 00-thesis-meta
 - phd-research-questions-and-roadmap
 - unified-variable-and-construct-glossary
 - wiki-sources-provenance-map

- ▼ 01-theory-psychology
 - assertiveness-st-as
 - conflict-resolution-st-rc
 - emotional-regulation-st-ri
 - emotional-sincerity-st-si
 - empathy-st-em
 - group-normative-context
 - socioemotional-competencies-st-x
 - sociometric-status-categories
 - teamwork-st-te

Wiki: Teoría GNN / Metodología

- > 00-thesis-meta
- > 01-theory-psychology
- > 02-theory-gnn
- > 03-methodology
- > 04-experiments-code
- > 05-papers-publications
- > 06-hypotheses
- index
- log

- ▼ 02-theory-gnn
 - bayesian-personalized-ranking-bpr
 - cs224w-graph-machine-learning-foundations
 - graphgym-modular-framework
 - over-smoothing-in-small-graphs
 - relational-gcn-architecture
 - signed-graph-neural-networks
 - spectral-graph-wavelets-glwt

- ▼ 03-methodology
 - composite-socioemotional-index-z-ise
 - graph-dirichlet-energy
 - habilmind-sociometric-survey-protocol
 - socioemotional-mismatch-index-delta
 - sociometric-relations-so1-so6
 - top-k-evaluation-metrics

PAL Programing Aided Language

> 00-thesis-meta
> 01-theory-psychology
> 02-theory-gnn
> 03-methodology
> 04-experiments-code
> 05-papers-publications
> 06-hypotheses
index
log

▼ 04-experiments-code
figure1-recall-distribution-boxplot
figure2-r2-dimension-histogram
repo-profiles-3-1-to-3-14
table4-classical-ml-baselines
table5-ablation-grid
table6-bootstrap-confidence-intervals
table7-linear-probe-r2-alignment

repo-profiles-3-1-to-3-14

Propiedades

≡ title	Thesis PAL Repositories Profiles (Repos 3.1 to 3.14)
≡ type	Reference
≡ topic	04-experiments-code
🔗 tags	experiments × pal × repositories × codebase ×
📄 sources	[{"id":"pal-index","resource":"./docs/INDEX/04_REPOSITORIES_PAL_INDEX.md","title":"..."}]
⋮ theoretical_ten...	Reconciling early exploratory scripts with production GraphGym architectures. ×
⋮ boundary_con...	Contains the complete 14-repository research progression. ×
📄 generated	{"by":"xgnn-vault-ingest/claude-code","at":"2026-08-23T12:00:00Z"}
≡ description	Summary catalog linking the 14 research repositories to their thesis roles and attested computations.
+ Añadir propiedad	

24

Thesis PAL Repositories Profiles (Repos 3.1 to 3.14)

Summary

Summary catalog linking the 14 research repositories to their thesis roles and attested computations. Full details are maintained in [docs/INDEX/04_REPOSITORIES_PAL_INDEX.md](#).
[[^]pal-index]

--

Core Progression Table

- Repo 3.1 (01-EDA-ST_x & ML): ST_x feature correlations ($r_{\max} = 0.347$).
- Repo 3.2 (02-GNN-v0): Inverse formulation collapse ($SO_x \rightarrow ST_x$).
- Repo 3.3 (03_Graph_Data_Quality): Universal intra-classroom variance proof.
- Repo 3.4 (04_Graph_Training-Zscore-PCT): Continuous node regression limits ($R^2 < 0.026$).
- Repo 3.5 (05_RandomForest-TopN_cada_S0x): Sociometric status (PS/IS) and classical ML baselines.
- Repo 3.6 (06_GLWT-LineaBaseRF-RGCN): Over-smoothing hazard in small graphs ($0.825 \rightarrow 0.502$).
- Repo 3.7/3.10 (10_Individual_Graph-ST_x): Controlled class-balance experiment.
- Repo 3.8 (08_GNN_Graph_Clustering): Macro topological clustering vs micro signals.
- Repo 3.9 (09_Individual_Graph_Analysis): Louvain community detection & centrality manifolds.
- Repo 3.11 (11-07_Data_Segmentation-EDA): PCA formulation of composite index Z_{ISE} .
- Repo 3.12 (12_GNN_LinkPrediction): STATEMENT_SO+ & birth of the Top-K ranking paradigm.
- Repo 3.13 (SDK-GNN-2026): ★ Production SDK for Paper 1.
- Repo 3.14 (60-Loyola-gym): ★ GraphGym modular research framework.

Theoretical Tensions

- Reconciling early exploratory scripts with production GraphGym architectures.

Boundary Conditions

- Contains the complete 14-repository research progression.

See Also

- [Table 5 Ablation Grid](#)
- [Table 4 Classical ML Baselines](#)
- [Paper 1 Draft Synthesis](#)

Nodos atractores: index.md

- > 00-thesis-meta
- > 01-theory-psychology
- > 02-theory-gnn
- > 03-methodology
- > 04-experiments-code
- > 05-papers-publications
- > 06-hypotheses
- index
- log

index

Propiedades

☰ okf_version 0.2

+ Añadir propiedad

4

5 Knowledge Bundle Index

6

7 00 Thesis Meta

8

- 9 • [PhD Research Questions & Dissertation Roadmap](#) - This doctoral research investigates how Relational Graph Neural Networks
- 10 • [Unified Variable and Construct Glossary](#) - This document establishes the single source of truth for all variable

11

12 01 Theory Psychology

13

- 14 • [Assertiveness \(ST_AS\)](#) - The ability to express opinions, feelings, and boundaries clearly, confidently,
- 15 • [Conflict Resolution & Guilt Recognition \(ST_RC\)](#) - Skills in negotiation, non-violent dispute mediation, recognizing one's
- 16 • [Interpersonal Emotional Regulation & Responsibility \(ST_RI\)](#) - The capacity to manage and modulate emotional impulses, endure frustration
- 17 • [Emotional Sincerity & Authenticity \(ST_SI\)](#) - Honesty in expressing internal emotional states, genuineness in peer
- 18 • [Empathy \(ST_EM\)](#) - The ability to understand and share the feelings of another, recognize
- 19 • [Group Normative Context & Classroom Moderation](#) - Classroom peer networks are not neutral aggregations of individual traits;
- 20 • [Socioemotional Competencies Framework \(CASEL & Bisquerra\)](#) - Socioemotional Learning (SEL) frameworks conceptualize the human capacities

✓ 02 Theory Gnn

- [Bayesian Personalized Ranking \(BPR\) for Graph Recommendation](#) - Pairwise BPR optimizes the relative ranking between positive chosen peers
- [Stanford CS224w Graph Machine Learning Foundations](#) - Anchored in the canonical Stanford CS224w curriculum, Graph Machine Learning
- [GraphGym Modular Experimentation Framework](#) - Based on Stanford GraphGym, this architectural design space standardizes
- [Over-Smoothing Hazard in Small Classroom Graphs](#) - In small, dense networks like school classrooms ($N \approx 25-30$),
- [Relational Graph Convolutional Networks \(R-GCN\) & Basis Decomposition](#) - R-GCN extends graph convolutions to multi-relational graphs by assigning
- [Signed Graph Neural Networks & Structural Balance Theory](#) - Signed Graph Neural Networks incorporate Heider's Structural Balance
- [Spectral Graph Wavelets \(GLWT\) & Multiscale Decomposition](#) - Graph Logarithmic Wavelet Transforms (GLWT) provide a localized, multiscale

03 Methodology

- [Composite Socioemotional Index \(Z_ISE\)](#) - Z_{ISE} synthesizes the 6 individual Habilmind socioemotional competencies
- [Graph Dirichlet Energy & Trait Homophily Testing](#) - Graph Dirichlet Energy measures the smoothness or spatial variation of
- [Habilmind Sociometric Survey Protocol & Dataset Specifications](#) - The dataset originates from Habilmind's computerized *Socioemociograma*
- [Socioemotional Mismatch Index \(Delta = Z_ISE - Z_SO\)](#) - The Socioemotional Mismatch Metric *Delta* quantifies the discrepancy
- [Sociometric Relations Taxonomy \(SO1 to SO6\)](#) - The sociometric instrument captures 6 directed relational channels covering
- [Top-K Recommendation & Ranking Evaluation Metrics](#) - Because each student selects exactly $K = 3$ academic peers (SO1), evaluation

04 Experiments Code

- [Paper 1 Figure 1: Per-Classroom Recall@3 Distribution Boxplot](#) - Figure 1 displays the boxplot distributions of Recall@3 across all 361
- [Paper 1 Figure 2: R2 Alignment Distribution Across 64 Dimensions](#) - Figure 2 plots the

Nodos atractores: log.md

- > 00-thesis-meta
- > 01-theory-psychology
- > 02-theory-gnn
- > 03-methodology
- > 04-experiments-code
- > 05-papers-publications
- > 06-hypotheses
- index
- log

log

OKF v0.2 Knowledge Vault Operations Log

This log records all automated scaffolding, compilation, attestation, and maintenance operations.

[2026-08-23 20:27:45] OKF Vault Bootstrap & Initial Multi-Corpus Compilation

- Actor: Antigravity AI Pair Programmer
- Schema: Open Knowledge Format (OKF v0.2)
- Actions Executed:
 1. Scaffolding: Initialized 7 topic directories (00-thesis-meta to 06-hypotheses) and configured .obsidian/ (app.json, core-plugins.json, graph.json with 7 decimal RGB color groups).
 2. Thesis Manifest: Deployed wiki/thesis.yaml defining doctoral boundaries and core paper sequences.
 3. Domain Ingestion: Compiled 10 psychometric & SEL concepts across CASEL/Bisquerra frameworks, Moreno sociometry, and individual ST x profiles.
 4. GNN Theory Ingestion: Compiled 7 graph machine learning theory concepts anchored in Stanford CS224w, R-GCN basis decomposition, BPR ranking, GLWT wavelets, and GraphGym.
 5. Methodology Compilation: Compiled 6 measurement and metric notes (SO1-SO6, Delta-mismatch, Dirichlet energy, Z_ISE, Top-K metrics).
 6. PAL Attestation Binding: Formulated 7 OKF Section 10 Attested Computation concepts binding Table 3, Table 4, Table 5, Table 6, Table 7, Figure 1, and Figure 2 to executable scripts in SDK-GNN-2026 and 60-loyola-gym.
 7. Publications & Hypotheses: Compiled Paper 1 synthesis, Claim Ledger D-01..D-25, and initial abductive hypotheses (H-XGNN-01, H-XGNN-02, H-XGNN-03).
- Integrity Status: 32 verified concept notes compiled with 100% valid YAML frontmatter and standard relative links.

Auditoría de Investigación Asistida por IA



Paper1 v2.5

Socioemotional Competencies as Determinants of Peer Choice in School Classrooms: Evidence from a Relational Graph Neural Network Study

Author names — anonymised for review

Abstract

A fundamental question in educational psychology is whether intra-individual socioemotional competencies determine inter-individual relational behavior. **Classical sociometric methods describe observed peer-choice patterns but cannot quantify the predictive weight of individual-level traits on collective relational structure.** We address this question empirically using a graph-based predictive framework applied to school classroom sociograms. We represent each classroom as a directed multigraph in which nodes are students annotated with six socioemotional competencies from the Habilmind psychopedagogical assessment (empathy ST_EM, assertiveness ST_AS, interpersonal emotional regulation ST_RI, teamwork ST_TE, conflict resolution ST_RC, emotional sincerity ST_SI), and edges encode six directed sociometric nomination types spanning academic, social, and leisure contexts with both positive (SO1-SO3) and negative (SO4-SO6) valences. To quantify the predictive influence of socioemotional competencies on SOx, we train a Relational Graph Convolutional Network (R-GCN) that encodes contextual affinity relations while preserving their semantic distinction via relation-specific weight matrices, and optimizes student embeddings via Bayesian Personalized Ranking (BPR) against academic peer-choice labels. This architecture isolates the contribution of socioemotional node features from purely structural signals, enabling controlled ablations. We evaluate on 361 classroom graphs comprising 8,982 students. **Introducing socioemotional node features raises Recall@3 for academic peer-choice prediction from 0.180 (structure-only baseline) to 0.462, a 157% gain, establishing that socioemotional competencies carry substantial predictive signal beyond structural position alone.** Post-hoc linear regression on the learned 64-dimensional embedding space reveals moderate alignment with the socioemotional competency variables (mean $R^2 = 0.182$, range 0.035–0.260), indicating that the model implicitly organizes its latent geometry according to socioemotional profiles despite being trained solely on peer-choice data. These two convergent analyses collectively **quantify a significant but partial influence of intra-individual socioemotional competencies** on inter-individual relational behavior,

consistent with developmental psychology literature showing that peer integration depends on group dynamics beyond individual trait profiles.

Keywords: sociometry, socioemotional learning, graph neural networks, Bayesian personalized ranking, peer choice, school classrooms, relational graph convolutional networks

1 Introduction

How students choose their peers in school classrooms is not a random process. Sociometric research has consistently shown that peer nominations, who students select as academic partners, who they enjoy spending time with, and who they prefer to play with, follow structured patterns that reflect both individual characteristics and group-level dynamics (Rubin et al., 1998; Cillessen and Bukowski, 2000; McPherson et al., 2001). The long-standing theoretical question is: to what extent are these inter-individual relational choices *determined* by intra-individual psychological traits?

Socioemotional learning (SEL) research proposes that competencies such as empathy, emotional regulation, and social initiative are the foundation of positive peer relations (Denham, 1998; CASEL, 2020). Yet the empirical evidence linking specific, measurable socioemotional competencies to specific, observable peer-choice patterns is surprisingly thin. Classical sociometric methods, based on nomination surveys administered to complete classrooms, yield rich descriptive maps of relational structure, but they are inherently observational: they describe *who chose whom* without providing a principled quantitative framework for explaining *why*.

Several methodological limitations compound this gap. First, studies that correlate socioemotional variables with sociometric status (e.g., number of nominations received) conflate many distinct relational processes into a single scalar (Sentse et al., 2010). Second, multi-context sociometric data, where students are asked to nominate peers across different social scenarios (academic work, class interaction, leisure play), is rarely analyzed jointly with intra-individual profiles. Third, the small, closed-group nature of classrooms makes standard correlation analyses underpowered for capturing the complex, non-linear, multi-hop relational patterns that may mediate socioemotional-behavioral links.

We argue that graph neural networks (GNNs) offer a principled measurement instrument for quantifying the predictive influence of socioemotional competencies on sociometric choices. Unlike summary statistics (e.g., in-degree, sociometric status), GNNs can integrate the full multi-relational structure of a classroom network; unlike classical regression, they can capture non-linear, multi-hop dependencies between a student's socioemotional profile and the profile of students who are (or are not) nominated.

paper1-forensic-audit-and-methodological-remediation.md

> 01-theory-psychology
> 02-theory-gnn
> 03-methodology
> 04-experiments-code
v 05-papers-publications
 paper1-claim-ledger-and-prov...
 paper1-critical-review-referee-r...
 paper1-forensic-audit-and-met...
 paper1-jcr-referee-report-revisi...
 paper1-socioemotional-rgcn-d...
 paper2-signed-gnns-roadmap
 paper3-xgmn-interventions-roa...
> 06-hypotheses
index
log

Paper 1 Forensic Audit & Methodological Remediation Ledger

Summary

Authoritative forensic audit ledger synthesizing the identification, root-cause diagnostics, mathematical remediation, and code-level verification of subtle methodological failure modes, evaluation sieves, and structural data leakages across Paper 1. [[^]editorial-diffs-log] [[^]bpr-code-audit-log] [[^]ethics-audit-doc] [[^]paper1-draft]

1. Executive Summary & Epistemic Significance for the Doctoral Dissertation

In applied Graph Machine Learning (GML), standard benchmark pipelines frequently report impressive empirical lifts on small, dense, or multi-relational graphs without subjecting their evaluation protocols to rigorous forensic scrutiny. During the peer-review revision and audit of Paper 1 (*Socioemotional Competencies as Determinants of Peer Choice in School Classrooms*), systematic investigation revealed seven subtle methodological, mathematical, and code-level failure modes.

Crucially, while external peer reviewers (GraphGym and Senior Educational AI referees) detected several high-level discrepancies, **both reviewers completely overlooked the direct structural target degree leakage in Ablation D** (where fixed node features $\mathbf{x}_v$ retained target in/out-degrees during cross-relational transfer).

Ingesting and formalizing this forensic ledger transforms Paper 1 from a routine application of Relational Graph Convolutional Networks (R-GCN) into an exemplary methodological case study on:

1. **Symmetry breaking vs. trait signal separation** in graph neural networks.
2. **Evaluation denominator sieves** in fixed-choice sociological nomination censuses.
3. **Data and feature leakage defense** in multi-relational edge ranking systems.
4. **Epistemological integration** between modern deep geometric learning and 30 years of parametric network modeling (ERGM, SAOM).

05-papers-publications

- paper1-claim-ledger-and-provenance
- paper1-critical-review-referee-report
- paper1-forensic-audit-and-methodological...
- paper1-jcr-referee-report-revision-plan
- paper1-socioemotional-rgcn-draft**
- paper2-signed-gnns-roadmap
- paper3-xgcn-interventions-roadmap

Paper 1: Socioemotional Competencies as Determinants of Peer Choice in School Classrooms

Summary

Paper 1 formulates the task of predicting academic work-partner choice (*SO1*) from in-class affinity (*SO2*) and leisure companionship (*SO3*) as a relational Top-3 ranking problem via R-GCN and Bayesian Personalized Ranking.^[^paper1-pdf]

Core Empirical Assertions

- Incremental Predictive Gain ($B3 \rightarrow Full$):** Adding 6-dimensional socioemotional competencies (ST_z) over a matched topological structural baseline ($B3$) yields a statistically significant but moderate improvement in Recall@3 of +7.4% ($\Delta = +0.031$, 95% CI [+0.018, +0.044], $p < 0.001$). The apparent +157% raw difference against the feature-free constant encoder ($A1, x = 1$) is decomposed into 65% degree signal ($A1 \rightarrow A3$, +0.182) and 11% net socioemotional signal ($B3 \rightarrow Full$).
- Superiority over Classical ML:** Outperforms 27D Random Forest and Logistic Regression baselines (0.234–0.240) by +92.5% (Table 4).
- Latent Manifold Alignment:** 64-dimensional linear probe regressions demonstrate distributed socioemotional encoding (Mean $R^2 = 0.182$, Table 7).
- Socioemotional Mismatch Concentration:** Model residuals concentrate heavily in $\Delta = Z_{ISB} - Z_{SO}$ outliers (§4.2).

Ethics, Data Governance & Compliance (JECR/RAISE Standard)

- **Institutional IRB Approval:** Protocol CEI-2024-XGNN approved by Universidad Loyola Andalucía Ethics Committee (15 March 2024).
- **Minor Protection & Consent:** Dual consent framework under Art. 7 LOPDGD 3/2018 (parental informed consent for <14, student consent + parental notification for ≥14, student assent).
- **Negative Nomination Safeguards:** Protected private administration, sequential positive-first queries, zero peer feedback, and non-harm empirical evidence from Spanish cohorts (González-Moreno et al., 2026^[?]).

Paper1 v2.6 - internal

Socioemotional Competencies as Determinants of Peer Choice in School Classrooms: Evidence from a Relational Graph Neural Network Study

Author names — anonymised for review

Abstract

A fundamental question in educational psychology is whether intra-individual socioemotional competencies determine inter-individual relational behavior. Classical sociometric methods describe observed peer-choice patterns using established frameworks like Exponential Random Graph Models (ERGMs) (??) and Stochastic Actor-Oriented Models (SAOMs) (??). We address this question empirically using a graph-based predictive framework applied to school classroom sociograms. We represent each classroom as a directed multigraph in which nodes are students annotated with six socioemotional competencies from the Habilmind psychopedagogical assessment (empathy ST_EM, assertiveness ST_AS, interpersonal emotional regulation ST_RI, teamwork ST_TE, conflict resolution ST_RC, emotional sincerity ST_SI), and edges encode six directed sociometric nomination types spanning academic, social, and leisure contexts with both positive (SO1–SO3) and negative (SO4–SO6) valences. To quantify the predictive influence of socioemotional competencies on SOx, we train a Relational Graph Convolutional Network (R-GCN) that encodes contextual affinity relations while preserving their semantic distinction via relation-specific weight matrices, and optimizes student embeddings via Bayesian Personalized Ranking (BPR) against academic peer-choice labels. This architecture isolates the contribution of socioemotional node features from purely structural signals, enabling controlled ablations. We evaluate on 361 classroom graphs comprising 8,982 students. Introducing socioemotional node features raises Recall@3 for academic peer-choice prediction from 0.431 (structural representation) to 0.462, a 7.4% relative gain ($\Delta = +0.031$), establishing that socioemotional competencies carry significant predictive signal beyond structural network position alone. Post-hoc linear regression on the learned 64-dimensional embedding space reveals moderate alignment with the socioemotional competency variables (mean $R^2 = 0.182$, range 0.035–0.260), indicating that the model implicitly organizes its latent geometry according to socioemotional profiles despite being trained solely on peer-choice data.

These two convergent analyses collectively quantify a **significant** but **partial influence of intra-individual socioemotional competencies** on inter-individual relational behavior, consistent with developmental psychology literature showing that peer integration depends on group dynamics beyond individual trait profiles.

Keywords: sociometry, socioemotional learning, graph neural networks, Bayesian personalized ranking, peer choice, school classrooms, relational graph convolutional networks

1 Introduction

How students choose their peers in school classrooms is not a random process. Sociometric research has consistently shown that peer nominations, who students select as academic partners, who they enjoy spending time with, and who they prefer to play with, follow structured patterns that reflect both individual characteristics and group-level dynamics (Rubin et al., 1998; Cillessen and Bukowski, 2000; McPherson et al., 2001). The long-standing theoretical question is: to what extent are these inter-individual relational choices *determined* by intra-individual psychological traits?

Socioemotional learning (SEL) research proposes that competencies such as empathy, emotional regulation, and social initiative are the foundation of positive peer relations (Denham, 1998; CASEL, 2020). Yet the empirical evidence linking specific, measurable socioemotional competencies to specific, observable peer-choice patterns is surprisingly thin. Classical sociometric methods, based on nomination surveys administered to complete classrooms, yield rich descriptive maps of relational structure, but they are inherently observational: they describe *who chose whom* without providing a principled quantitative framework for explaining *why*.

Several methodological limitations compound this gap. First, studies that correlate socioemotional variables with sociometric status (e.g., number of nominations received) conflate many distinct relational processes into a single scalar (Sentse et al., 2010). Second, multi-context sociometric data, where students are asked to nominate peers across different social scenarios (academic work, class interaction, leisure play), is rarely analyzed jointly with intra-individual profiles. Third, the small, closed-group nature of classrooms makes standard correlation analyses underpowered for capturing the complex, non-linear, multi-hop relational patterns that may mediate socioemotional-behavioral links.

We argue that graph neural networks (GNNs) offer a principled measurement instrument for quantifying the predictive influence of socioemotional competencies on sociometric choices. Unlike summary statistics (e.g., in-degree, sociometric status), GNNs can integrate the full multi-relational structure of a classroom network; unlike classical regression, they



El Generador de Hipótesis



El Generador de Hipótesis

06-hypotheses

h-xgnn-01-signed-structural-balance

h-xgnn-02-hyperbolic-hierarchy-embeddings

h-xgnn-03-xgnn-delta-outlier-interventions

hypothesis-generation-protocol

index

log

33

34 Hypothesis H-XGNN-01: Signed Balance Channels for 35 Polarization Recovery

36 Formal Hypothesis Statement

37 Decomposing negative sociometric nominations (SO_{ext} — SO_6) into balanced (+, −) and
(−, −) dual-channel signed attention convolutions will transform negative tie noise into
predictive signal, boosting Recall@3 on polarized classrooms where isotropic message
passing fails.

38

39 Theoretical Premise & Rationale

40 In Paper 1 Ablation C1, adding negative relations naively into standard R-GCN yielded
negligible gain (0.467 vs 0.462) due to negative edge cancellation. Applying Heiderian
signed convolutions ($W^+h^+ + W^-h^-$) preserves structural opposition boundaries.

41

42 Theoretical Tensions

- 43 • Resolves the tension between negative edge degradation in isotropic GNNs and
structural balance theory.

44

45 Boundary Conditions

- 46 • Applies to classrooms with high negative nomination density ($|E_{neg}| / |E_{pos}| > 0.4$).

47

48 See Also

- 49 • [Signed Graph Neural Networks](#)
- 50 • [Paper 2 Signed GNNs Roadmap](#)
- 51 • [Table 5 Ablation Grid](#)

52

53 paper1-draft Research Corpus, Paper 1 Ablation C Noise Finding.

54

